

GORAKHPUR, India

WMO: 423790

Lat: 26.740N

Lon: 83.450E

Elev: 259

StdP: 14.56

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	45.7	47.9	35.4	30.6	76.0	38.8	35.0	67.4	9.1	66.5	7.3	63.6	0.5	270	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	15.5	105.5	73.2	102.9	73.2	100.3	73.5	84.5	91.5	83.0	91.0	82.2	90.4	5.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.8	178.8	87.0	81.1	163.4	86.9	80.1	157.6	87.0	49.3	90.7	47.5	91.6	46.6	90.4	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	8.8	6.9	DB	41.4	107.4	2.9	2.5	39.3	109.3	37.6	110.8	36.0	112.2	33.9	114.0	(4)
(5)			WB	39.5	84.9	3.2	1.7	37.2	86.1	35.3	87.1	33.5	88.0	31.2	89.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	77.9	58.7	66.9	76.3	85.7	88.5	89.0	85.6	85.4	84.4	80.2	71.8	62.4	(6)
(7)		DBStd	10.89	5.02	4.34	4.87	4.09	4.06	4.49	2.96	2.30	2.78	3.66	3.44	4.85	(7)
(8)		HDD50	6	5	0	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD65	329	198	25	0	0	0	0	0	0	0	0	1	105	(9)
(10)		CDD50	10199	275	473	814	1072	1194	1169	1103	1096	1031	936	653	383	(10)
(11)		CDD65	5048	4	78	349	622	729	719	638	631	581	471	203	23	(11)
(12)		CDH74	63265	20	537	3557	8596	10755	10569	8289	8075	7058	4446	1227	136	(12)
(13)	CDH80	32608	0	81	1676	5207	6617	6350	3900	3677	3042	1763	291	4	(13)	
(14)	Wind	WSAvg	2.0	1.3	1.8	2.5	2.9	3.1	2.9	2.4	2.4	2.0	1.0	0.6	0.8	(14)
(15)	Precipitation	PrecAvg	42.9	0.7	0.7	0.4	0.2	1.9	5.5	11.4	10.9	7.8	2.0	0.1	0.2	(15)
(16)		PrecMax	76.8	2.8	2.5	2.4	0.7	5.9	19.5	27.1	24.3	20.6	9.3	2.0	2.1	(16)
(17)		PrecMin	24.6	0.0	0.0	0.1	0.0	0.1	0.0	1.6	3.3	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	12.6	0.7	0.7	0.5	0.2	1.5	5.1	6.5	5.3	4.4	2.8	0.4	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.6	86.2	98.1	106.0	109.1	108.5	97.3	95.3	95.0	93.5	87.7	80.2	(19)
MCWB			61.3	67.2	69.1	69.6	71.9	75.1	81.4	82.9	80.2	77.3	69.2	63.7	(20)	
2%		DB	73.9	82.5	94.5	103.5	105.7	105.6	94.8	93.5	93.4	91.6	85.8	77.5	(21)	
		MCWB	60.1	65.7	68.3	68.6	72.5	76.4	81.5	82.4	80.2	77.0	68.8	62.5	(22)	
5%		DB	71.6	79.9	92.0	101.2	103.0	102.2	93.0	92.1	92.0	89.9	83.6	75.3	(23)	
		MCWB	59.3	64.4	67.5	68.9	73.4	77.8	81.4	81.9	80.0	76.3	68.2	62.0	(24)	
(25)	10%	DB	68.9	77.4	89.2	98.7	100.2	98.6	91.4	90.8	90.5	88.3	81.2	72.3	(25)	
(26)		MCWB	58.4	63.3	67.2	69.5	74.5	79.3	81.2	81.5	79.8	75.6	67.4	60.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	71.1	75.9	78.3	82.3	84.5	86.0	87.5	85.8	81.7	73.8	67.4	(27)
MCD B			73.2	81.4	90.7	95.4	98.2	96.3	89.6	91.9	88.3	88.6	81.7	75.8	(28)	
2%		WB	62.4	68.7	72.5	76.6	80.7	83.4	83.6	85.0	82.3	80.3	71.7	64.8	(29)	
		MCD B	70.6	79.6	87.4	94.2	95.9	95.2	89.6	89.1	89.2	87.9	81.1	73.9	(30)	
5%		WB	60.9	66.4	70.3	75.2	79.4	82.3	82.5	82.8	81.3	79.1	70.3	63.1	(31)	
		MCD B	68.7	76.7	85.9	92.5	94.3	93.2	89.8	89.2	88.7	86.3	79.6	71.8	(32)	
10%		WB	59.1	64.5	68.7	73.8	78.3	81.4	81.8	81.9	80.6	78.0	68.9	61.7	(33)	
		MCD B	67.1	74.4	84.5	90.8	92.8	92.0	89.0	88.7	88.4	85.0	78.2	70.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.3	20.9	24.1	25.4	21.1	15.5	9.3	9.7	10.9	16.7	22.0	19.0	(35)
MCDBR			23.0	23.1	27.1	28.4	26.0	22.4	11.9	11.8	13.2	17.6	23.1	23.5	(36)	
MCWBR			12.0	10.2	8.8	9.0	8.2	6.8	4.2	5.0	4.4	6.4	9.6	11.4	(37)	
5% WB		MCDBR	20.2	21.1	24.5	23.4	19.6	16.0	10.9	11.1	11.7	14.0	19.9	20.6	(38)	
		MCWBR	11.5	11.1	10.2	9.5	8.0	6.0	5.2	5.7	5.3	6.4	9.2	11.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.561	0.535	0.528	0.611	0.699	0.719	0.633	0.583	0.574	0.625	0.657	0.620	(40)	
taud		1.725	1.832	1.878	1.740	1.612	1.609	1.803	1.949	1.938	1.749	1.615	1.637	(41)		
Ebn at Noon		210	233	246	229	210	204	222	233	231	207	183	186	(42)		
Edh at Noon		65	62	63	74	84	84	69	59	58	67	72	69	(43)		
(44)	All-Sky Solar Radiation	RadAvg	927	1383	1819	2059	2020	1723	1419	1456	1448	1438	1168	916	(44)	
(45)		RadStd	132	100	60	96	114	160	128	84	104	93	62	114	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page